SUPERTEAM IRELAND SUBMISSION

MinKYC

IDEATHON @ DOGPATCH LABS
FRIDAY MARCH 13, 2026

Direct Reference Links:

Website: https://getoutofthatgarden.github.io/minkyc-website/

Repository: https://github.com/GetOutOfThatGarden/MinKYC

Hackathon Submission Note

This is my official submission for the Superteam Ireland Ideathon. While I cannot be physically at Dogpatch Labs today, I am actively participating in this guided ideation session to build a concrete foundation for the **Solana Colosseum Hackathon** starting in April. I am grateful for any feedback offered, as it will directly assist in refining the project for the global competition.

MinKYC is the privacy-first compliance layer for Solana Mobile. I eliminate KYC data breach liability by proving identity with on-device Zero-Knowledge cryptography—designed specifically for the Solana Seeker.

The "Toxic Asset" Problem

In the current internet landscape, your personal data is a toxic asset. Storing passports and IDs turns servers into high-value targets for hackers, creating a multi-million dollar liability for every platform.

CASE STUDY: THE 2025 DISCORD SCANDAL

In September 2025, Discord suffered a catastrophic data breach via a third-party support vendor (5CA/Zendesk). **Over 2.1 million government ID photos** were allegedly stolen, affecting users who had simply appealed for age verification. Discord confirmed that at least 70,000 highly sensitive documents were exposed.

The Lesson: If you hold the data, you will eventually lose the data. Platforms can no longer afford the "Trust Us" model of identity.

Immense Financial Costs

The average cost of a data breach in 2025 hit **$4.88 Million** globally. For U.S.-based entities, that cost skyrockets to over **$10 Million** per incident, including fines, forensic audits, and legal settlements.

The Regulatory Trap

New laws like the UK's Online Safety Act and Europe's GDPR mandate strict age verification but impose crippling penalties for data mismanagement. Platforms are caught between compliance and liability.

How It Works: Technical Deep-Dive

MinKYC moves the verification process to the "edge"—directly onto the user's mobile device—leveraging **Noir ZK circuits** and the hardware enclaves of modern smartphones.

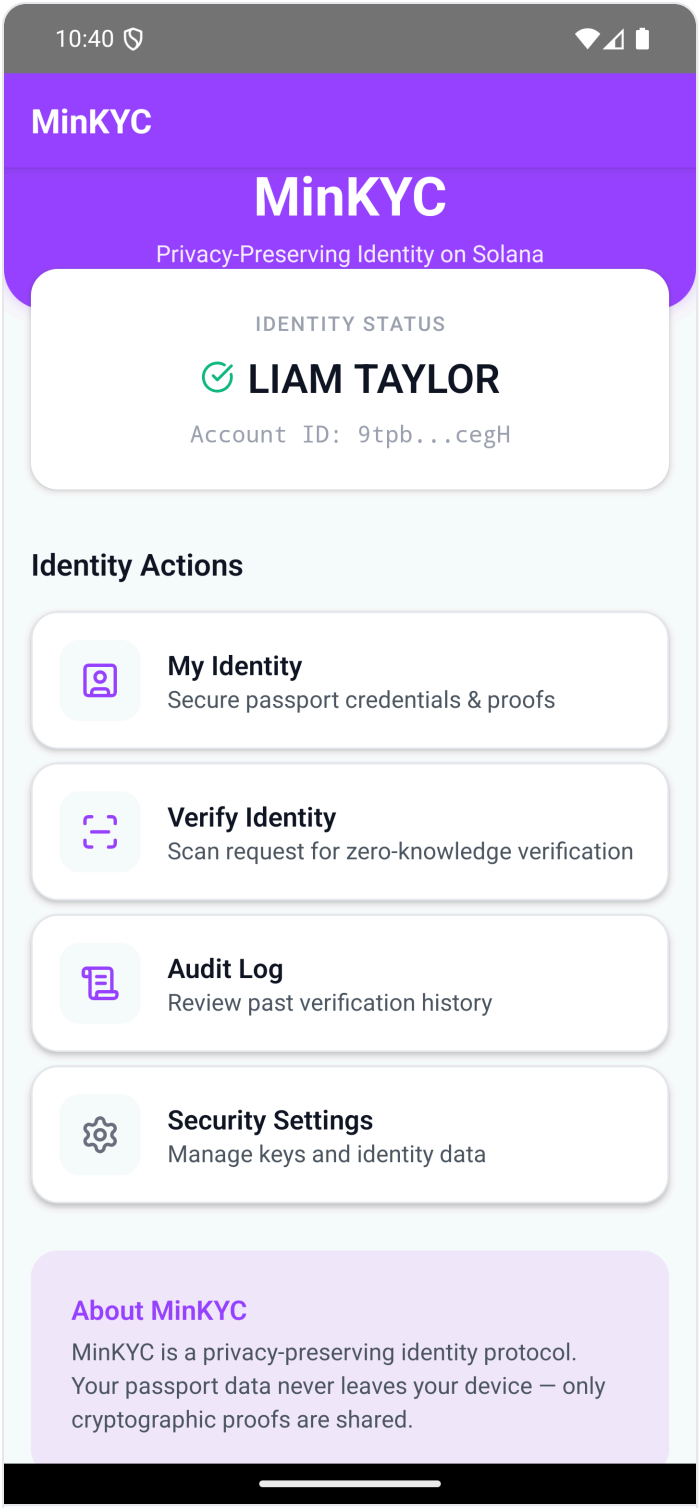
The Noir ZK Pipeline

> **Hardware Extraction:** The mobile app uses NFC to scan the user's biometric passport. The data is held only in the device's secure memory—never written to disk or sent to a server.

> **Circuit Generation:** A Noir ZK circuit processes the signed passport data. It mathematically confirms that the signature matches a trusted government authority and that the "Birth Date" field satisfies a condition (e.g., age >= 18).

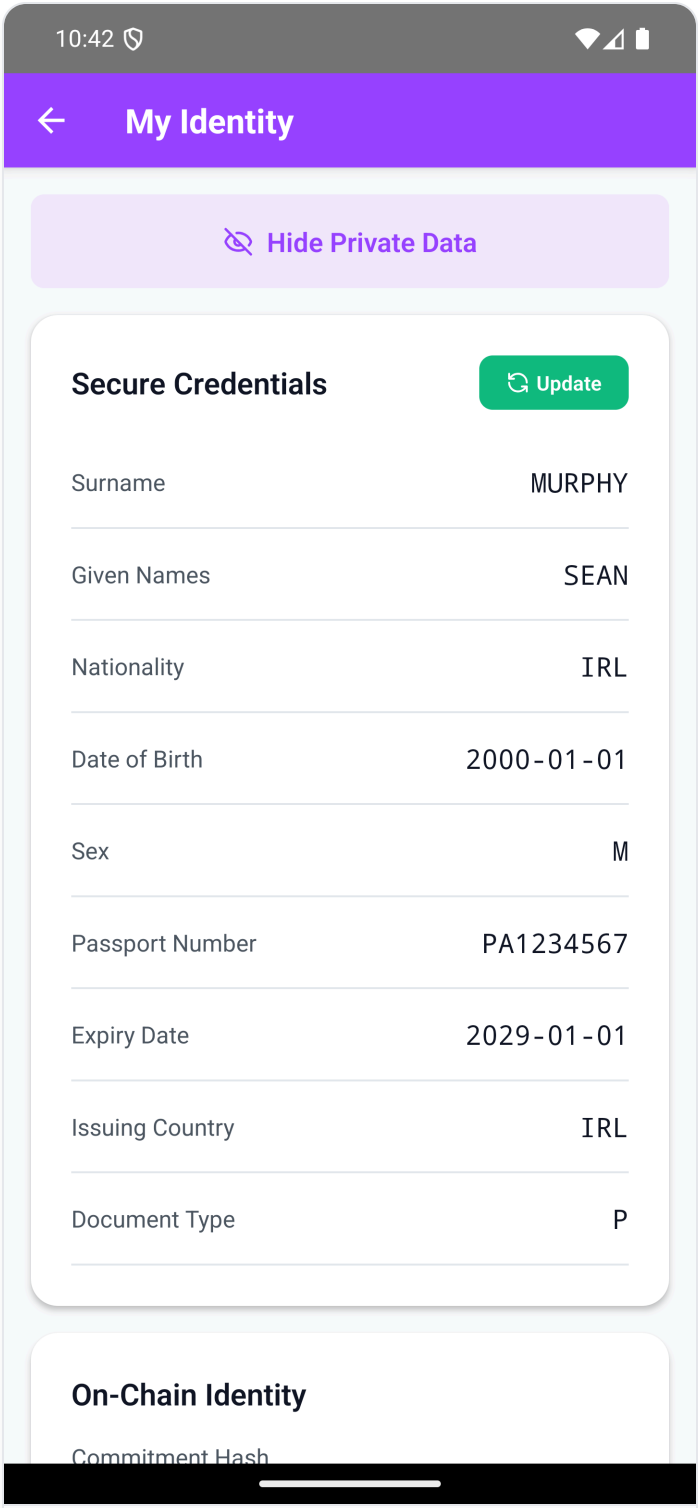
> **Zero-Knowledge Export:** The app exports a cryptographic proof of the result. This proof contains **zero PII**. It is a mathematical string that says "This claim is true" without revealing the underlying data.

> **On-Chain Verification:** The Solana smart contract (or any dApp via Mobile Wallet Adapter) receives this proof. It verifies the proof on-chain in milliseconds for a fraction of a cent.



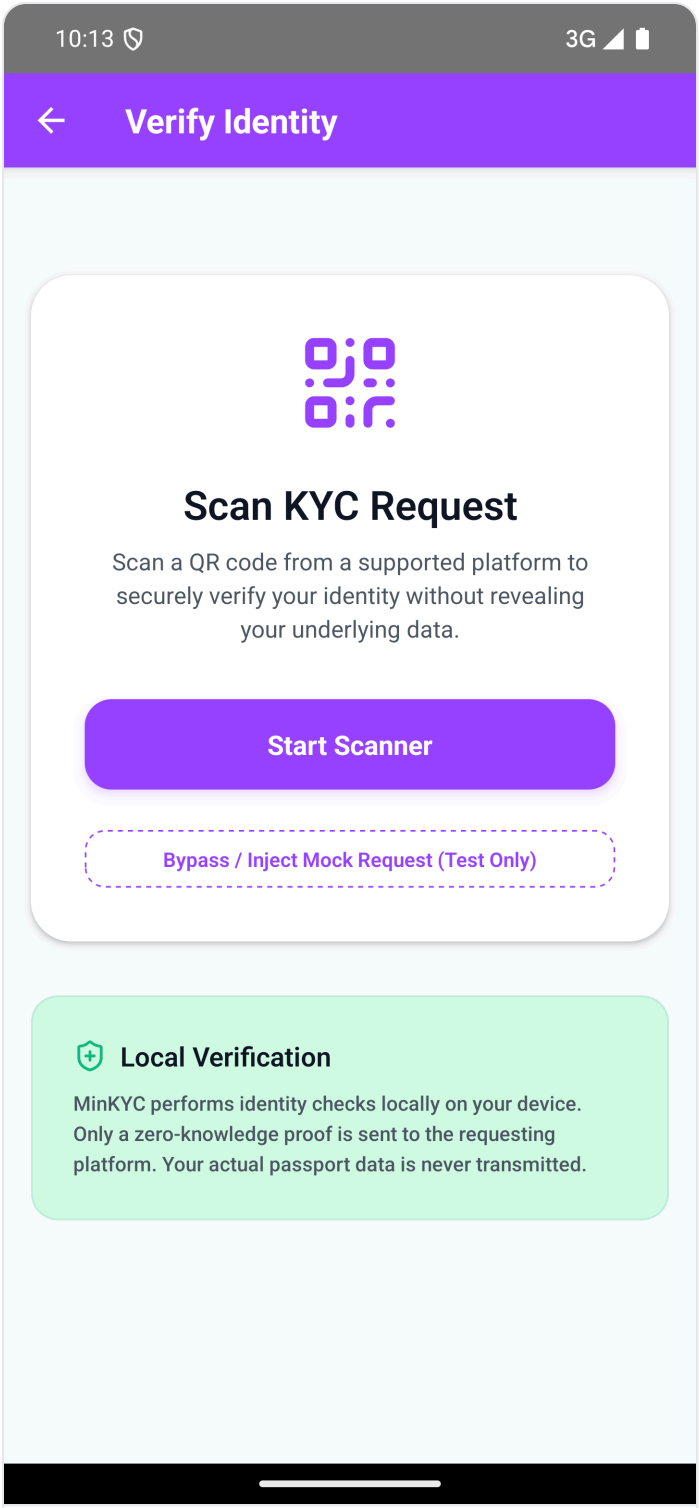
Native Home Interface

A clean dashboard displaying current identity status and quick actions, built for the Seeker screen.



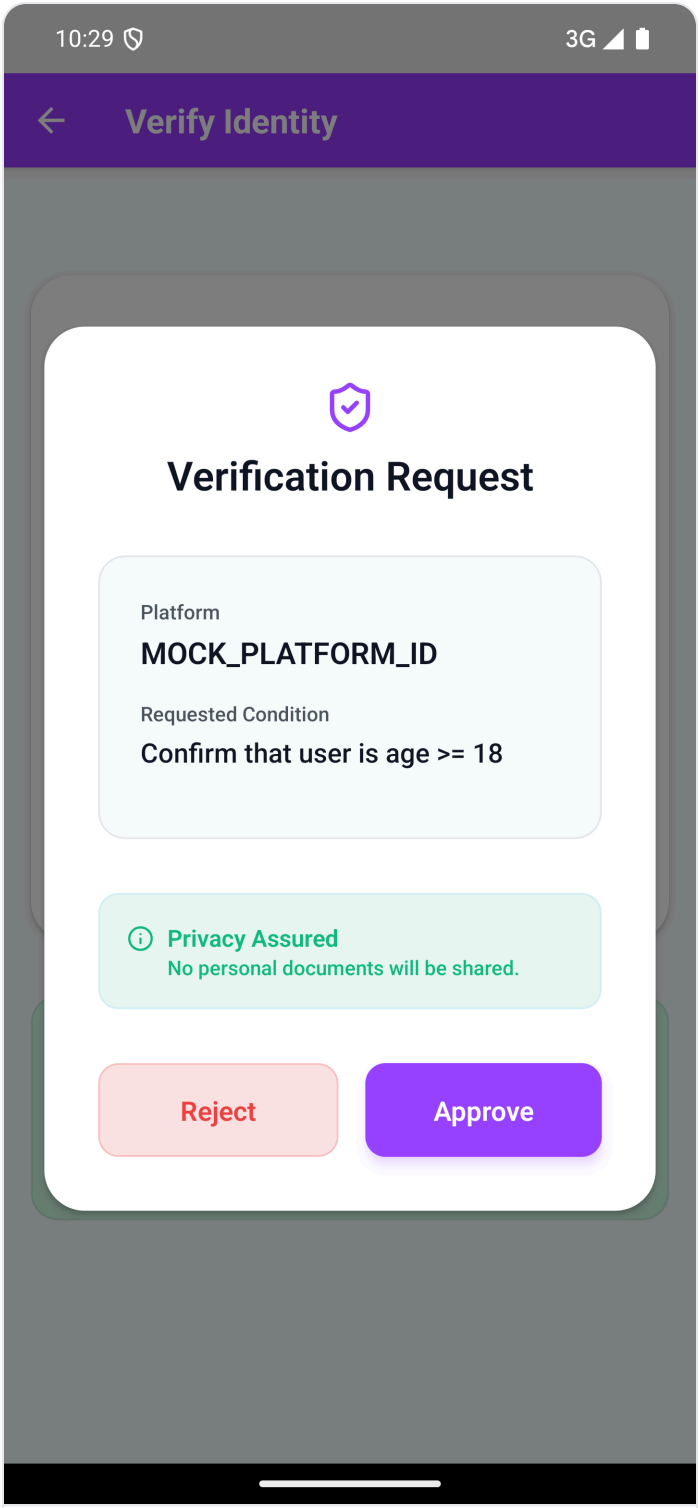
Encrypted Identity Vault

User credentials are encrypted and stored locally. Data never leaves the device.



Platform Bridge Scanner

Users scan a QR from a dApp to initiate a seamless mobile-to-web verification handshake.



The ZK Proof Request

The user is prompted to generate a specific proof (e.g., Age > 18) without revealing their birthdate.

Go-To-Market (GTM) Strategy

I am targeting the most compliance-sensitive verticals in the Solana ecosystem first, where the cost of verification is currently high and the risk of centralization is a deterrent.

Phase 1: High-Compliance DeFi & Prediction Markets

Integrating with Solana protocols that require "Accredited Investor" status or geographic fencing (e.g., Parcl, Drift). MinKYC allows them to stay compliant without becoming a "Bank".

Phase 2: The Seeker Ecosystem

By partnering with Solana Mobile, MinKYC aims to be the default identity provider for the dApp Store. Seeker owners get a "Verified Human" badge across all apps without exposing their data.

Phase 3: Web2 Pivot

Transitioning to traditional age-gated retail and online services (gaming, social media) that are reeling from the Discord-style breaches.

Possible Pricing Strategies

Transactional (Pay-per-Proof)

Platforms pay **$0.50 per successful ZK verification**. This replaces the expensive $2.00+ fees charged by traditional OCR services like Persona or Jumio.

"The Compliance Shield"

A yearly subscription for platforms that offloads all data-handling liability. For a fixed fee, I provide the ZK-infrastructure and the platforms never touch an ID photo again.

Why Solana?

To replace traditional APIs, KYC verifications must be instant and cheap. Solana's high throughput and sub-dollar fees allow for sub-second ZK proof verification and the minting of immutable audit receipts, creating a transparent yet private compliance trail.

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